

A Philosophical-Mathematical Representation of Deductive and Inductive Logic: A Diagrammatic and Formal Analysis

Hsuan Fu Wang
Independent researcher
hsuanfuwang@gmail.com

Abstract

We give a diagrammatic representation of deductive and inductive reasoning in which deduction is a certain passage from thought to existence, $\varphi \rightarrow \exists$, and induction is a tentative, iterated passage from the study of existence to the theory of thought, $s\exists \odot t\varphi$. We make the diagram precise in a possible-worlds semantics: thoughts are propositions about worlds, deduction is semantic entailment, polarity $\uparrow, \downarrow$ distinguishes affirmation from modus tollens, and an inductive theory is supported by a study when some world is consistent with both. Within this setting we prove that deduction is sound and transitive, that iterated observation can only narrow the supported theories, that a single counterexample falsifies a universal theory, and that no finite study entails a universal theory over an unobserved individual. The last statement is a formal version of Hume's problem of induction. As an application we realize the framework in quantum information theory: thoughts become subspaces, deduction becomes certain measurement and unitary evolution, induction becomes state tomography, and the classical setting is recovered as the case of commuting observables. The quantum model reproduces refinement, falsification and underdetermination, and adds a limitation with no classical counterpart: measurements in a single basis never determine a quantum state. We further give a Bayesian refinement of support, under which every genuine test a theory predicts confirms it, a ranked semantics for defeasible inference, and an order on levels under which iteration moves upward. Every theorem of the paper is verified in the Lean 4 proof assistant with Mathlib.

Keywords: deductive logic; inductive logic; diagrammatic reasoning; possible-worlds semantics; Bayesian confirmation; problem of induction; quantum logic; quantum state tomography; Lean theorem prover.

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1 Introduction

Deductive logic derives particular conclusions from general premises with certainty: all men are mortal; Socrates is a man; therefore Socrates is mortal [1]. Inductive logic builds general theories from particular observations with only tentative support: many swans have been seen to be white, so perhaps all swans are white. Textual arguments and static diagrams, such as Venn diagrams and Peirce’s existential graphs [28], show the structure of a single inference well but do not show reasoning as a process in which deduction and induction alternate.

This paper proposes a diagram (Figure 1) that places both modes of reasoning in one picture, with a straight arrow for deduction, a circular arrow for induction, and a foundation of iterations, hypotheses, levels, binary states and polarities. We then give the diagram a precise semantics and prove what it asserts.

Related work. Diagrams as rigorous logical representations go back to Peirce and have been given formal semantics, for example for Venn diagrams by Shin [31]; our diagram differs in depicting two modes of inference and their interaction rather than the content of a single argument. The semantics of Section 3 is the standard possible-worlds reading of propositions [22]. Our qualitative notion of support is the consistency requirement underlying confirmation theory [15], and its quantitative refinement in Section 5.3 is Bayesian incremental confirmation [7, 12, 13]. Falsification follows Popper [29], and the treatment of defeasible inference in Section 5.4 uses the preferential semantics of Kraus, Lehmann and Magidor [21, 26]. The quantum model draws on quantum logic [6, 8] and quantum state tomography [17, 27]. Machine-checked formalization of philosophical arguments has been pursued with automated and interactive provers [4, 14]; we use Lean 4 and Mathlib [9, 32].

Contributions.

1. A diagrammatic framework for deduction $\varphi \rightarrow \exists$ and induction $s\exists \circ t\varphi$ (Section 2).
2. A possible-worlds semantics for every symbol of the diagram (Section 3).
3. Theorems for each mode: soundness, transitivity and modus tollens for deduction (Section 4); monotone refinement, falsification and the failure of induction as deduction (Theorem 5.4) for induction (Section 5); and a truth-value characterization of the foundation (Section 6).
4. An application to quantum information theory (Section 7): a dictionary from the framework to quantum states, measurements and Hamiltonians; quantum counterparts of the deductive and inductive theorems (Propositions 7.1 and 7.5); the classical framework as the commuting case (Proposition 7.3); and physical implementations, a thermodynamic reading and experimental proposals.
5. Three extensions: Bayesian confirmation as the quantitative inductive polarity (Section 5.3), defeasible inference (Section 5.4), and ordered levels (Section 6.1).
6. A machine-checked formalization of every theorem of Sections 4–7 in Lean 4 [9] and Mathlib [32] (Appendix A).

Outline. Section 2 presents the diagram and Section 3 its semantics. Sections 4 to 6 prove the results for deduction, induction and the foundation, including the extensions. Section 7 develops the quantum model, Section 8 discusses the results and their limitations, and Appendix A maps every result to its Lean declaration.

2 The diagrammatic framework

The diagram has two primary rows and a foundation (Figure 1).

- **Deduction**, $\varphi \rightarrow \exists$. A thought φ (premises) leads to an existence $\exists$ (a conclusion) along a straight arrow, read *hence*. The passage is certain. Polarity $\uparrow$ marks an affirmed conclusion and $\downarrow$ a negated one.
- **Induction**, $s\exists \circ t\varphi$. The study of existence $s\exists$ (observations) leads to a theory of thought $t\varphi$ (a hypothesis) along a circular arrow, read *conclude*. The passage is tentative and is repeated as new observations arrive.
- **Foundation**. Iterations i , hypotheses h and levels l , together with binary states $0, 1$ and polarities $\uparrow, \downarrow$, generate the thoughts used above.

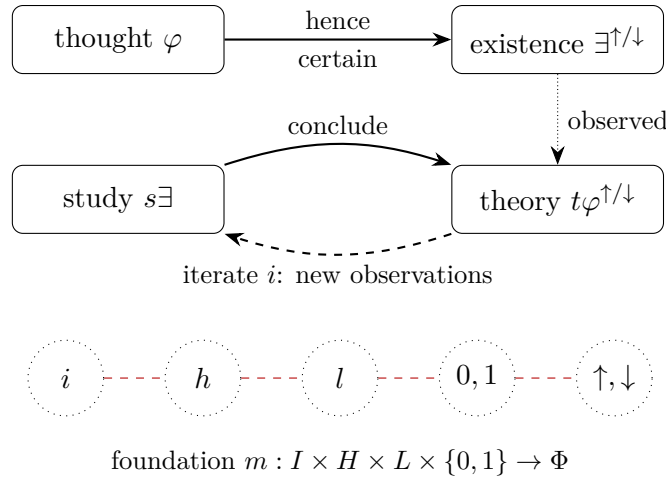


Figure 1: Deduction (top row), induction (middle row) and the foundation (bottom).

Table 1 fixes the reading of each symbol. The symbol $\exists$ is kept for the diagram only; the set of possible existences is written $\mathcal{W}$, so that it is not confused with the existential quantifier.

Symbol	Informal meaning	Formal reading (Section 3)
φ	thought	proposition $\varphi : \mathcal{W} \rightarrow \text{Prop}$
$\exists$	existence	proposition about worlds; $\mathcal{W}$ is the set of worlds
s	study	list of observations
t	theory	proposition about worlds
$\rightarrow$	hence	semantic entailment $\models$
$\circ$	conclude	support (Definition 3.4)
i	iteration	adding an observation to a study
h, l	hypothesis, level	indices of the foundation map m
$0, 1$	binary state	truth values
$\uparrow, \downarrow$	polarity	affirmation, negation

Table 1: Symbols of the framework.

3 Formal setting

Fix a set $\mathcal{W}$ of *possible worlds*. Every symbol of the diagram is read as follows.

Definition 3.1 (Thought). A *thought* is a proposition about worlds, that is, a map $\varphi : \mathcal{W} \rightarrow \text{Prop}$. We write Φ for the set of thoughts and $\varphi(w)$ for “ φ holds in w ”. An existence is likewise a thought; the distinction between φ and $\exists$ in the diagram is one of role (premise versus conclusion), not of type.

Definition 3.2 (Entailment). A thought φ *entails* ψ , written $\varphi \models \psi$, if $\{w \in \mathcal{W} : \varphi(w)\} \subseteq \{w \in \mathcal{W} : \psi(w)\}$.

Entailment is a relation, not a function: a thought entails many conclusions, so no map $\Phi \rightarrow \exists$ could assign to each thought its single conclusion.

Definition 3.3 (Polarity). The polarities are $\uparrow$ and $\downarrow$. For a proposition p , set $p^\uparrow = p$ and $p^\downarrow = \neg p$.

Definition 3.4 (Study and support). A *study* is a finite list $s = (o_1, \dots, o_n)$ of thoughts, the observations. A theory t is *supported* by s , written $s \circ t$, if some world satisfies t and every o_k :

$$s \circ t \quad :\Longleftrightarrow \quad \exists w \in \mathcal{W}, t(w) \wedge \bigwedge_{k=1}^n o_k(w).$$

Remark 3.5. A probabilistic reading of the circular arrow would require $P(t \mid s) > 0$. That quantity is undefined when the study has probability zero. If $\mathcal{W}$ is finite and P gives every world positive probability, then $P(t \wedge s) > 0$ holds exactly when $s \circ t$, so [Definition 3.4](#) is the qualitative content of the probabilistic condition and needs no choice of prior. (The same holds for countable $\mathcal{W}$; the Lean proof treats the finite case.) A quantitative refinement is given in [Section 5.3](#).
Prior.prob_and_pos_iff_supported

4 Deduction

Proposition 4.1 (Material implication). For propositions p, q , $(p \rightarrow q) \iff (\neg p \vee q)$.
imp_iff_not_or

Proof. If $p \rightarrow q$, then either p holds and so q holds, or $\neg p$ holds; this case split uses the law of excluded middle. Conversely, if $\neg p \vee q$ and p , the first disjunct is impossible, so q . $\square$

Theorem 4.2 (Deduction is certain). If $\varphi \models \psi$ and $\varphi(w)$, then $\psi(w)$. deduction_sound

Proof. By [Definition 3.2](#), w lies in the set of worlds satisfying φ , which is contained in the set satisfying ψ . $\square$

Proposition 4.3 (Chains of deduction). If $\varphi \models \psi$ and $\psi \models \chi$, then $\varphi \models \chi$. Entails.trans

Proof. Set inclusion is transitive. $\square$

Proposition 4.4 (Negative polarity is modus tollens). If $\varphi \models \psi$ and $\psi(w)^\downarrow$, then $\varphi(w)^\downarrow$.
modus_tollens

Proof. If $\varphi(w)$ held, [Theorem 4.2](#) would give $\psi(w)$, contradicting $\neg\psi(w)$. $\square$

Example 4.5 (Deductive swan). Let B be a set of birds with predicates Swan and White. If $\forall b, \text{Swan}(b) \rightarrow \text{White}(b)$ and $\text{Swan}(a)$, then $\text{White}(a)$. In the diagram this is $\varphi \rightarrow \exists^\uparrow$. swan_syllogism

5 Induction

5.1 Iterative refinement

Iteration i adds an observation to a study. The following results say that iteration can only narrow the set of supported theories, which is the precise sense in which induction refines.

Proposition 5.1. *If $(o, o_1, \dots, o_n) \circ t$, then $(o_1, \dots, o_n) \circ t$.* *Supported.of_cons*

Theorem 5.2 (Refinement is monotone). *If every observation of s occurs in s' and $s' \circ t$, then $s \circ t$.* *Supported.mono*

Proof. A world witnessing $s' \circ t$ satisfies t and every observation of s' , hence every observation of s . [Proposition 5.1](#) is the case $s' = (o) \dashv\vdash s$. □

5.2 Falsification and the problem of induction

For the swan examples take worlds to be colourings $c : B \rightarrow \{\text{true}, \text{false}\}$ recording which birds are white. For a finite list *seen* of birds define

$$\text{SeenWhite}_{\text{seen}}(c) :\iff \forall x \in \text{seen}, c(x) = \text{true}, \quad \text{AllWhite}(c) :\iff \forall x \in B, c(x) = \text{true}.$$

Proposition 5.3 (Falsification). *If $\text{Swan}(b)$ and $\neg \text{White}(b)$, then $\neg \forall x, \text{Swan}(x) \rightarrow \text{White}(x)$.* *falsification*

Proof. Instantiate the universal statement at b . □

Falsification is deductive: a single observation settles the matter, in line with Popper [29]. Confirmation is not, however many observations are made.

Theorem 5.4 (Induction is not deduction). *Let B have decidable equality, let *seen* be a finite list of birds, and let $b \in B$ with $b \notin \text{seen}$. Then $\text{SeenWhite}_{\text{seen}} \not\models \text{AllWhite}$.* *induction_not_deduction*

Proof. Define $c(x) = \text{false}$ if $x = b$ and $c(x) = \text{true}$ otherwise. Every $x \in \text{seen}$ differs from b , so $c(x) = \text{true}$ and $\text{SeenWhite}_{\text{seen}}(c)$ holds. But $c(b) = \text{false}$, so $\text{AllWhite}(c)$ fails. Hence the worlds satisfying $\text{SeenWhite}_{\text{seen}}$ are not contained in those satisfying AllWhite . □

Proposition 5.5 (Inductive swan). *For every finite list *seen*, $(\text{SeenWhite}_{\text{seen}}) \circ \text{AllWhite}$.* *allWhite_supported*

Proof. The colouring that makes every bird white satisfies both. □

Together, [Theorem 5.4](#) and [Proposition 5.5](#) state the asymmetry the diagram depicts: the theory “all swans are white” remains supported by every finite study, so it may be held along the circular arrow, but no finite study entails it, so it never travels the straight arrow. This is a formal version of Hume’s problem of induction [18].

5.3 Quantitative support: Bayesian confirmation

Support is qualitative: it asks only whether a theory survives the study. The inductive polarity $t\varphi^\uparrow$ suggests more, an “ascending probability”: the study should make the theory more probable. This is the notion of incremental confirmation in Bayesian confirmation theory [7, 12, 13]. We make it precise for a finite set of worlds.

Definition 5.6 (Prior and confirmation). Let $\mathcal{W}$ be finite. A *prior* is a weight $p : \mathcal{W} \rightarrow (0, \infty)$; it defines $P(\varphi) = \sum_{\varphi(w)} p(w) / \sum_w p(w)$ and $P(t \mid s) = P(t \wedge s) / P(s)$, where a study s is read as the conjunction of its observations. The study *confirms* t , written $t\varphi^\uparrow$, if $P(t \mid s) > P(t)$, and *disconfirms* it, $t\varphi^\downarrow$, if $P(t \mid s) < P(t)$. Confirms, Disconfirms

Proposition 5.7 (Confirmation implies support). *If s confirms t , then $s \circ t$.* Confirms.supported

Proof. $P(t \mid s) > P(t) \geq 0$, so $P(t \wedge s) > 0$, and Remark 3.5 applies. □

Theorem 5.8 (Hypothetico-deductive confirmation). *Suppose t is satisfiable, $t \models s$, and some world falsifies s . Then s confirms t .* confirms_of_entails

Proof. Since $t \models s$, $P(t \wedge s) = P(t) > 0$, so $P(t \mid s) = P(t) / P(s)$. A world falsifying s has positive weight, so $P(s) < 1$ and $P(t) / P(s) > P(t)$. □

Theorem 5.8 joins the two arrows: a theory gains probability from a study exactly of the kind it deductively predicts, provided the study was a genuine test (it could have failed).

Proposition 5.9 (Iteration raises confirmation). *If t is satisfiable, $t \models s'$ and $s' \models s$, then $P(t \mid s) \leq P(t \mid s')$.* condProb_mono

Proof. Both posteriors equal $P(t)$ divided by the probability of the study, and $P(s') \leq P(s)$. □

Proposition 5.10 (Falsification, quantitatively). *If $s \models \neg t$ and t is satisfiable, then $P(t \mid s) = 0 < P(t)$, so s disconfirms t .* Disconfirms.of_refutes

Corollary 5.11 (Bayesian swan). *Let B be finite and fix any prior on colourings. For every non-empty list $seen$, $(\text{SeenWhite}_{seen})$ confirms AllWhite; and if every bird of $seen$ occurs in $seen'$, then $P(\text{AllWhite} \mid \text{SeenWhite}_{seen}) \leq P(\text{AllWhite} \mid \text{SeenWhite}_{seen'})$.* allWhite_confirmed, allWhite_condProb_mono

Proof. AllWhite $\models \text{SeenWhite}_{seen}$, the all-white colouring satisfies AllWhite, and the all-black colouring falsifies SeenWhite_{seen} ; apply Theorem 5.8 and Proposition 5.9. □

With Theorem 5.4, Corollary 5.11 gives the full inductive picture: white sightings raise the probability of “all swans are white” at every step, yet never entail it, while one black swan sends its probability to 0 (Proposition 5.10). This is the ascending probability of the circular arrow.

5.4 Defeasible inference

Between certain deduction and probabilistic confirmation lies defeasible inference: conclusions drawn from several premises that hold normally but may be withdrawn. We use the ranked semantics of Kraus, Lehmann and Magidor [21].

Definition 5.12 (Defeasible entailment). A *ranking* is a map $r : \mathcal{W} \rightarrow \mathbb{N}$, lower ranks being more normal. Premises $\Gamma = (\gamma_1, \dots, \gamma_n)$ *defeasibly entail* ψ , written $\Gamma \mid \sim \psi$, if ψ holds in every r -minimal world satisfying all γ_k . Normally, NormallyFrom

Theorem 5.13 (Preferential reasoning). *Defeasible entailment contains entailment ($\varphi \models \psi$ implies $\varphi \mid \sim \psi$) and satisfies right weakening, AND, OR, CUT and cautious monotonicity: if $\varphi \mid \sim \psi$ and $\varphi \mid \sim \chi$ then $\varphi \wedge \psi \mid \sim \chi$.* Normally.of_entails, Normally.cautious_mono

Proof. All but cautious monotonicity are immediate from the definition. For cautious monotonicity let w be minimal among $\varphi \wedge \psi$ -worlds and v any φ -world. Since ranks are natural numbers there is a minimal φ -world m with $r(m) \leq r(v)$; it satisfies ψ , so $r(w) \leq r(m) \leq r(v)$. Thus w is minimal among φ -worlds and satisfies χ . □

Proposition 5.14 (Defeasible inference is nonmonotone). *With worlds $(\text{penguin}, \text{flies}) \in \{0, 1\}^2$ ranked so that flying non-penguins are most normal and flightless penguins next, a bird normally flies, $() \mid \sim \text{flies}$, but $(\text{penguin}) \not\mid \sim \text{flies}$.* normally_nonmonotone

By contrast, entailment is monotone in its premises (`entails_premises_mono`). Defeasible inference thus sits between the arrows: like deduction it is closed under the rules of [Theorem 5.13](#), and like induction it can be overturned by new information.

6 The foundation

Definition 6.1 (Foundation map). Let I, H, L be sets of iterations, hypotheses and levels, and let $\beta : I \times H \times L \rightarrow \Phi$ assign a base thought to each index. The *foundation map* $m : I \times H \times L \times \{0, 1\} \rightarrow \Phi$ is

$$m(i, h, l, 1) = \beta(i, h, l), \quad m(i, h, l, 0) = \neg\beta(i, h, l).$$

Thus state 1 carries polarity $\uparrow$ and state 0 carries polarity $\downarrow$. Since existences are thoughts ([Section 3](#)), m takes values in Φ .

Proposition 6.2 (The binary state is the truth value). *For every world w and $b \in \{0, 1\}$, $m(i, h, l, b)(w)$ holds if and only if $\beta(i, h, l)(w)$ agrees with b , that is, $\beta(i, h, l)(w) \iff b = 1$.* *m_spec*

Proof. For $b = 1$ both sides say $\beta(i, h, l)(w)$. For $b = 0$ both sides say $\neg\beta(i, h, l)(w)$. □

6.1 Ordered levels

The indices i, h, l are uninterpreted. We now order the levels, reading lower levels as observational and higher ones as theoretical.

Definition 6.3 (Hierarchy). Let L be preordered. A *hierarchy* assigns a thought β_l to each level so that $l \leq l'$ implies $\beta_{l'} \models \beta_l$: a theory entails its observable consequences. Given a study s , level l is *established* if $s \models \beta_l$ and *tenable* if $s \odot \beta_l$. Hierarchy

Theorem 6.4 (Iteration moves upward). *Established and tenable levels form down-sets. If every observation of s occurs in s' , every level established by s is established by s' , and every level tenable given s' is tenable given s . If s is satisfiable, every established level is tenable. Hierarchy.Established.mono*

Proof. Down-sets: entailment is transitive, and a world satisfying $\beta_{l'}$ satisfies β_l . A world satisfying s' satisfies s , so $s \models \beta_l$ gives $s' \models \beta_l$; the tenable case is [Theorem 5.2](#). A world satisfying s then satisfies every established β_l . □

Iteration therefore raises the floor of established levels and lowers the ceiling of tenable ones, with the floor below the ceiling as long as the study is consistent. Along an iteration $i \mapsto s_i$ that only adds observations, a level established at step i stays established at every later step ([Hierarchy.Established.iterate](#)).

7 Application: a quantum information model

This section gives a physical realization of the framework in quantum information theory. Thoughts become subspaces of a Hilbert space, deduction becomes certain measurement and unitary evolution, induction becomes quantum state tomography, and the binary and polarized foundation becomes qubits and spins. The classical setting of [Section 3](#) reappears as the special case in which all observables commute ([Proposition 7.3](#)). All propositions of this section are formalized in Lean with Mathlib ([Appendix A](#)); the physical implementations, Landauer’s principle and the experimental proposals are not.

Throughout, $\mathcal{H}$ is a finite-dimensional complex Hilbert space, a *state* is a unit vector $|v\rangle \in \mathcal{H}$ or a density operator ρ , and a *projector* is an operator P with $P = P^\dagger = P^2$. For projectors, $P \leq Q$ means $\text{ran } P \subseteq \text{ran } Q$, equivalently $QP = P$. Physical time is written τ , to keep t for theories.

7.1 Dictionary

7.2 Deduction as certain measurement

A thought φ is realized by a projector P_φ : the property “the system lies in $\text{ran } P_\varphi$ ”. Entailment is subspace inclusion, and negative polarity is the orthogonal complement, $P^\perp = I - P$.

Proposition 7.1 (Quantum deductive certainty). *Let $P_\varphi \leq P_\psi$ and let $|v\rangle$ be a state with $P_\varphi|v\rangle = |v\rangle$. Then measuring $\{P_\psi, I - P_\psi\}$ on $|v\rangle$ yields P_ψ with probability $\langle v|P_\psi|v\rangle = 1$. Moreover $I - P_\psi \leq I - P_\varphi$. quantum_deduction_sound*

Proof. Since $P_\psi P_\varphi = P_\varphi$, we get $P_\psi|v\rangle = P_\psi P_\varphi|v\rangle = P_\varphi|v\rangle = |v\rangle$, so $\langle v|P_\psi|v\rangle = \langle v|v\rangle = 1$. For the second claim, $(I - P_\varphi)(I - P_\psi) = I - P_\psi - P_\varphi + P_\varphi P_\psi = I - P_\psi$, using $P_\varphi P_\psi = (P_\psi P_\varphi)^\dagger = P_\varphi$. □

[Proposition 7.1](#) is the quantum form of [Theorem 4.2](#) and [Proposition 4.4](#). Unitary evolution is deductive in the same sense: $|v\rangle \mapsto U|v\rangle$ is deterministic.

Symbol	Quantum entity	Physical meaning
φ (thought)	projector P_φ , or a state $ v\rangle$	a property; a superposition of possibilities
$\exists$ (existence)	eigenstate $ n\rangle$	a definite measurement outcome
$\models$ (hence)	$P_\varphi \leq P_\psi$	subspace inclusion
s (study)	outcomes of measurements $\{P_n\}$	observed statistics
t (theory)	reconstructed state $\hat{\rho}$, or model $U(\theta)$	hypothesis about the system
i (iteration)	$U(\tau) = e^{-i\hat{H}\tau/\hbar}$, circuit layers, adaptive rounds	refinement in time
h (hypothesis)	measurement basis $\{ n\rangle\}$	what is asked of the system
l (level)	energy levels E_n of $\hat{H}$	hierarchical structure
0, 1 (binary)	qubit basis $ 0\rangle, 1\rangle$	computational basis
$\uparrow, \downarrow$ (polarity)	spin states $ \uparrow\rangle, \downarrow\rangle$; $P^\downarrow = I - P$	affirmation, negation

Table 2: Quantum realization of the symbols of Table 1.

Proposition 7.2 (Unitary deduction). *Let $U^\dagger U = I$. If $UP_\varphi U^\dagger \leq P_\psi$ and $P_\varphi|v\rangle = |v\rangle$, then $P_\psi U|v\rangle = U|v\rangle$. Moreover $\text{Tr}(U\rho U^\dagger U P U^\dagger) = \text{Tr}(\rho P)$ for all ρ, P .* *unitary_deduction, born_unitary_conj*

Proof. $U|v\rangle = UP_\varphi|v\rangle = (UP_\varphi U^\dagger)U|v\rangle$, and $P_\psi(UP_\varphi U^\dagger) = UP_\varphi U^\dagger$. The second claim is cyclicity of the trace. $\square$

A general measurement is not certain. For a projective measurement $\{P_n\}$ with $\sum_n P_n = I$ and $P_n P_m = \delta_{nm} P_n$, the Born rule gives

$$|v\rangle \mapsto \frac{P_n|v\rangle}{\|P_n|v\rangle\|} \quad \text{with probability} \quad p(n) = \langle v|P_n|v\rangle.$$

What is deduced from the state is the distribution p , not the individual outcome. For a spin, $|v\rangle = \alpha|\uparrow\rangle + \beta|\downarrow\rangle$ yields $|\uparrow\rangle$ with probability $|\alpha|^2$ and $|\downarrow\rangle$ with probability $|\beta|^2$; certainty is the case $|\alpha| = 1$ or $|\beta| = 1$.

Proposition 7.3 (The classical framework is the commuting case). *Let $\{|n\rangle\}_{n \in \mathcal{W}}$ be an orthonormal basis indexed by the finite set of worlds $\mathcal{W}$, and set $P_\varphi = \sum_{n: \varphi(n)} |n\rangle\langle n|$. Then*

1. $P_\varphi \leq P_\psi$ if and only if $\varphi \models \psi$, and $P_{\neg\varphi} = I - P_\varphi$; *classicalProj_le_iff, classicalProj_not*
2. if $\rho = \sum_n p_n |n\rangle\langle n|$ with every $p_n > 0$, then for a study $s = (o_1, \dots, o_k)$ and theory t , $\text{Tr}(\rho P_t P_{o_1} \cdots P_{o_k}) > 0$ if and only if $s \circlearrowleft t$. *classical_trace_pos_iff*

Proof. (1) $\text{ran } P_\varphi$ is spanned by the $|n\rangle$ with $\varphi(n)$, and such a basis vector lies in $\text{ran } P_\psi$ exactly when $\psi(n)$. The second claim holds because every basis vector satisfies exactly one of φ and $\neg\varphi$. (2) The projectors are diagonal, so their product is the projector onto the $|n\rangle$ with $t(n) \wedge o_1(n) \wedge \cdots \wedge o_k(n)$, and the trace is $\sum p_n$ over those n . A sum of positive numbers is positive exactly when it is non-empty, which is Definition 3.4. $\square$

Part (2) realizes Remark 3.5 physically. Outside the commuting case the projectors of $\mathcal{H}$ do not form a Boolean algebra [6, 33]. For a spin, with P_{x+} the projector onto $(|\uparrow\rangle + |\downarrow\rangle)/\sqrt{2}$,

$$P_{x+} \wedge (P_{z+} \vee P_{z-}) = P_{x+} \wedge I = P_{x+}, \quad (P_{x+} \wedge P_{z+}) \vee (P_{x+} \wedge P_{z-}) = 0,$$

so the distributive law fails.

Proposition 7.4 (Quantum logic is not distributive). *In the lattice of subspaces of $\mathbb{C}^2$, with $Z_{\pm}$ spanned by $|\uparrow\rangle, |\downarrow\rangle$ and X_{+} by $|\uparrow\rangle + |\downarrow\rangle$: $X_{+} \wedge (Z_{+} \vee Z_{-}) = X_{+} \neq 0$ while $(X_{+} \wedge Z_{+}) \vee (X_{+} \wedge Z_{-}) = 0$.*
quantum_not_distributive

Proof. $Z_{+} \vee Z_{-} = \mathbb{C}^2$. A vector $c(1, 1) = d(a, b)$ with $a \neq b$ forces $c = 0$, so both meets are 0. $\square$

This marks where the quantum model goes beyond [Section 4](#): the identity of [Proposition 4.1](#) relies on a Boolean structure that incompatible observables lack.

7.3 Induction as state tomography

In the quantum model a study is a record of outcomes $n_1, \dots, n_N$ obtained by measuring identical copies of an unknown state, and a theory is a state ρ that explains them. The likelihood of ρ is $L_N(\rho) = \prod_{k=1}^N \text{Tr}(\rho P_{n_k})$, and ρ is *supported* when $L_N(\rho) > 0$. Maximum-likelihood tomography [17] selects

$$\hat{\rho}_{\text{ML}} = \arg \max_{\rho} \prod_{k=1}^N \text{Tr}(\rho P_{n_k}),$$

and adaptive tomography chooses the next basis from the outcomes so far,

$$h_{k+1} = f(n_1, \dots, n_k; h_1, \dots, h_k).$$

This is the circular arrow $s \exists \circ t \varphi$ with iteration i indexing the rounds.

Proposition 7.5 (Quantum refinement, falsification and underdetermination).

1. *Each further outcome can only shrink the set of supported states: $L_{N+1}(\rho) > 0$ implies $L_N(\rho) > 0$.*
QSupported.of_concat
2. *Suppose N measurements of $\{|\uparrow\rangle\langle\uparrow|, |\downarrow\rangle\langle\downarrow|\}$ all give $\uparrow$. Then $|\uparrow\rangle$ is the maximum-likelihood state, yet every state $|\theta, \phi\rangle = \cos \theta |\uparrow\rangle + e^{i\phi} \sin \theta |\downarrow\rangle$ with $0 \leq \theta < \pi/2$ is supported, with likelihood $\cos^{2N} \theta > 0$.*
spinUp_maxLikelihood, spinState_supported
3. *A single outcome $\downarrow$ refutes the theory $|\uparrow\rangle$: its likelihood becomes 0.*
spinUp_refuted
4. *No number of measurements in this basis distinguishes $|\theta, \phi\rangle$ from $|\theta, \phi'\rangle$: their outcome statistics are equal for all ϕ, ϕ' .*
likelihood_spinState_phase

Proof. (1) $L_{N+1}(\rho) = L_N(\rho) \text{Tr}(\rho P_{n_{N+1}})$, and every factor $\text{Tr}(\rho P)$ lies in $[0, 1]$, so $0 \leq L_{N+1}(\rho) \leq L_N(\rho)$. (2) $\langle \theta, \phi | \uparrow \rangle \langle \uparrow | \theta, \phi \rangle = \cos^2 \theta$, so the likelihood is $\cos^{2N} \theta$, which is positive for $\theta < \pi/2$ and largest at $\theta = 0$. (3) $\langle \uparrow | \downarrow \rangle \langle \downarrow | \uparrow \rangle = 0$. (4) The outcome probabilities are $\cos^2 \theta$ and $\sin^2 \theta$, independent of ϕ . $\square$

Parts (1)–(3) are the quantum counterparts of [Theorem 5.2](#), [Theorem 5.4](#) and [Proposition 5.3](#). Part (4) is a limitation with no classical counterpart: data from one basis leave the phase undetermined however much of it is collected, so a tomographically complete set of bases is needed. In the language of the framework, the hypothesis h (the basis) bounds what any study can reveal.

7.4 Foundation: qubits and spins

The foundation map m of Section 6 is realized by the projector onto $|b\rangle$ in the basis selected by h : state 1 affirms and state 0 negates, as in Proposition 6.2. A register of n qubits with spin degrees of freedom has states

$$|v\rangle = \sum_{x \in \{0,1\}^n} \sum_{\sigma \in \{\uparrow, \downarrow\}^n} c_{x,\sigma} |x\rangle |\sigma\rangle,$$

and iteration is a circuit of depth d , $|v_{\text{final}}\rangle = U_d \cdots U_2 U_1 |v_{\text{initial}}\rangle$.

7.5 Physical implementations

Deduction: quantum gates. Single-qubit gates (X , Y , Z , Hadamard) and two-qubit gates (CNOT, CZ) act as $|v\rangle \mapsto U|v\rangle$, deterministic transformations that realize the straight arrow [11, 27].

Induction: variational algorithms. The variational quantum eigensolver and the quantum approximate optimization algorithm prepare $|v(\theta)\rangle = U(\theta)|0\rangle$ and update

$$\theta_{k+1} = \theta_k - \eta \nabla_{\theta} \langle v(\theta_k) | \hat{H} | v(\theta_k) \rangle,$$

with the gradient estimated from measurement outcomes. The parameters are the theory, the measurements are the study, and each update is one turn of the circular arrow [30].

Induction: quantum machine learning. Training data $\{(x_k, y_k)\}$ form the study, a parametrized circuit $U(\theta)$ is the theory, and training minimizes $L(\theta) = \sum_k |y_k - \langle v(x_k) | M | v(x_k) \rangle|^2$ for an observable M [5].

7.6 Thermodynamic reading

Information has a physical cost. Recording a study of N binary outcomes uses N bits, and by Landauer's principle resetting that record for reuse dissipates at least $Nk_B T \ln 2$ of heat [3, 23]. The two arrows then differ thermodynamically:

- *Deduction preserves entropy.* Unitary evolution leaves the von Neumann entropy $S(\rho) = -\text{Tr} \rho \ln \rho = -\sum_k \lambda_k \ln \lambda_k$ unchanged, $S(U\rho U^\dagger) = S(\rho)$, because $U\rho U^\dagger$ and ρ have the same characteristic polynomial; no information is gained or lost. `vonNeumannEntropy_unitary_conj`
- *Induction lowers the observer's entropy, at a price.* Observation reduces the expected Shannon entropy of the observer's belief about the theory, $H(T | S) \leq H(T)$. As with Maxwell's demon, the reduction is paid for by the Landauer cost of erasing the record [3].

7.7 Noise

Real devices are noisy. A simple model is the depolarizing channel $\mathcal{D}_\varepsilon(\rho) = (1 - \varepsilon)\rho + \varepsilon I/d$ on a d -dimensional system.

Proposition 7.6 (Deduction and falsification under noise). *Let $0 \leq \varepsilon$. If $\text{Tr}(\rho P) = 1$ for a projector P , then $\text{Tr}(\mathcal{D}_\varepsilon(\rho)P) \geq 1 - \varepsilon$. For a spin and $\varepsilon > 0$, $\text{Tr}(\mathcal{D}_\varepsilon(|\uparrow\rangle\langle\uparrow|) |\downarrow\rangle\langle\downarrow|) = \varepsilon/2 > 0$.*
`born_depolarize_ge, noisy_spinUp_not_refuted`

Proof. $\text{Tr}(\mathcal{D}_\varepsilon(\rho)P) = (1 - \varepsilon) \text{Tr}(\rho P) + \varepsilon \text{Tr}(P)/d$ and $\text{Tr} P \geq 0$. For the spin, $\text{Tr}(|\uparrow\rangle\langle\uparrow| |\downarrow\rangle\langle\downarrow|) = 0$ and $\text{Tr}(|\downarrow\rangle\langle\downarrow|)/2 = 1/2$. $\square$

Noise thus weakens both arrows in a controlled way: deductive certainty degrades by at most the noise rate, and falsification (Proposition 7.5(3)) becomes statistical, since a single outcome $\downarrow$ no longer refutes $|\uparrow\rangle$. Error correction is the means of restoring them.

7.8 Experimental proposals

1. *Tomography.* Prepare an unknown state $|v\rangle$, measure in a tomographically complete set of bases, reconstruct $\hat{\rho}$ by maximum likelihood, and report the fidelity $F = \langle v|\hat{\rho}|v\rangle$. Plotting F against N shows inductive refinement directly.
2. *Quantum walk.* The state $|v(\tau)\rangle = \sum_n \alpha_n(\tau)|n\rangle|c_n\rangle$, with position $|n\rangle$ and coin $|c_n\rangle$, displays iteration as coherent evolution.
3. *Error correction.* A logical qubit $|v_L\rangle$ is a protected thought, the noisy physical qubits are existence, the syndrome measurement is the study $s\exists$, and the decoder's inferred error is the theory $t\varphi$; applying the correction is then a deduction.

8 Discussion

Philosophy. The straight arrow carries Descartes' movement from thought to existence [10]: what is entailed holds in every world where the premises hold (Theorem 4.2). The circular arrow carries Hume's and Popper's view of induction: a generalization is supported, never entailed (Theorem 5.4), and can be overturned by a single counterexample (Proposition 5.3). Levels l in the foundation give a place for Kant's hierarchical structuring of thought [20].

Quantum analogies. The quantum model of Section 7 suggests three further analogies, which we state as analogies rather than results. *Complementarity:* as quantum objects show wave and particle aspects, reasoning involves both abstract thought (φ) and concrete existence ($\exists$). *Uncertainty:* incompatible observables cannot both be sharp, which is a physical limitation on what any study can reveal, distinct from Hume's logical one (Proposition 7.5(4)). *Entanglement:* entangled systems show correlations that no classical model reproduces, suggesting that logical relations between parts of a system need not reduce to relations between the parts; these correlations cannot be used to signal, so they do not conflict with causality. These views connect to the idea that physical processes are computations [11, 25].

Education and cognition. The top row of the diagram is suitable for a first course in logic; the foundation invites discussion of polarity and iteration. The visual layout follows the view that diagrams make inferential structure explicit [19, 24].

Limitations and future work.

- *Infinite sets of worlds.* The confirmation results of Section 5.3 assume finitely many worlds. For infinite or continuous sets of worlds the same definitions can be stated with measures and conditional measures, and we expect Theorem 5.8 and Proposition 5.9 to carry over for theories of positive prior probability. The sign of confirmation does not depend on the prior (Corollary 5.11); its size does.
- *Hypotheses and iterations.* Levels are now ordered (Section 6.1), but the hypothesis index h remains uninterpreted. A natural reading lets h index a partition of $\mathcal{W}$, the distinctions a

study under h can draw, so that a theory is identifiable under h only if it is a union of cells. [Proposition 7.5\(4\)](#) is the quantum instance.

- *Higher-order logic.* Every theorem is proved for an arbitrary type of worlds, including function spaces: in the swan examples a world is a predicate on birds, so AllWhite already quantifies over a predicate. A higher-order logic with its own syntax and proof system is not treated; Henkin semantics [16], with soundness proved against the possible-worlds reading, is the natural route.
- *Physical realization.* The quantum model is limited in practice by decoherence, state-preparation and measurement errors, and scale [2, 30]. [Proposition 7.6](#) quantifies the effect of depolarizing noise on both arrows. Showing that the three-qubit repetition code restores deductive certainty to order ε^2 is a next step; error-corrected implementations remain experimental work.
- *Thermodynamics.* Landauer’s principle is a physical law, assumed rather than derived. The inequality $H(T \mid S) \leq H(T)$ is a standard consequence of the concavity of entropy but is not formalized here.
- *Learners.* Whether the diagram helps learners is an empirical question. We propose a preregistered randomized trial: students in an introductory logic course receive a lesson built on the diagram or the same lesson in prose, and are tested on classifying arguments as deductive or inductive, on recognizing fallacies (affirming the consequent, hasty generalization) and on predicting the effect of a counterexample, with a retention test four weeks later. Detecting a medium effect ($d = 0.5$) with power 0.8 at $\alpha = 0.05$ requires about 64 students per arm.

9 Conclusion

We have given a diagram for deduction and induction, a possible-worlds semantics for its symbols, and machine-checked theorems for what it depicts: deduction is certain and transitive, negative polarity is modus tollens, iteration narrows the supported theories, one counterexample falsifies, and no finite study entails a universal theory. Bayesian confirmation makes the ascending probability of the circular arrow precise: a theory is confirmed by every genuine test it predicts, more so as the tests accumulate, yet never entailed. Defeasible inference sits between the two arrows, and ordered levels show iteration moving upward. Realized in quantum information theory, the same structure appears as certain measurement and unitary evolution on one side and state tomography on the other, with the classical framework as the commuting case and with a physical limit on induction that has no classical counterpart. This suggests that quantum mechanics offers a natural physical setting for the interplay of deduction and induction.

Statements and declarations

Code availability. The Lean 4 formalization (toolchain v4.34.1, Mathlib commit d13f23b7) is in the directory `lean` of the repository

<https://github.com/pupubear007/deductive-inductive-logic>

Appendix A lists the declaration proving each result.

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Competing interests. The author has no competing interests to declare.

Use of AI tools. The Lean formalization and parts of the text were prepared with the assistance of an AI coding assistant (Claude Code, Anthropic). All proofs are checked by the Lean kernel. The author takes full responsibility for the content of the paper.

A Lean formalization

All theorems of the paper are formalized in Lean 4 (toolchain v4.34.1) in the directory `lean/Wan gLogic/` of the repository given under Code availability. `Basic.lean` covers Sections 3–6 except the extensions below and uses only the core library. The other files use Mathlib [32] (commit d13f23b7): `Confirmation.lean` (Remark 3.5 and Section 5.3), `Defeasible.lean` (Section 5.4), `Levels.lean` (Section 6.1), `Quantum.lean` (Propositions 7.1, 7.3 and 7.5) and `QuantumExtensions.lean` (Propositions 7.2, 7.4 and 7.6 and entropy invariance). No file contains `sorry`, and every declaration depends at most on the axioms `propext`, `Quot.sound` and `Classical.choice`. The project builds with

```
cd lean && lake exe cache get && lake build
```

The file `lean/ComparatorChallenges/InductionNotDeduction.json` states Theorem 5.4 for independent checking with the Comparator tool (<https://github.com/leanprover/comparator>). Not formalized: Landauer’s principle, the inequality $H(T \mid S) \leq H(T)$, and the physical implementations and experimental proposals of Section 7.

Conventions. In Section 5.3 the set of worlds is finite and a prior is a strictly positive weight; probabilities are normalized sums of weights. In Section 5.4 a ranking takes values in $\mathbb{N}$. In Proposition 7.1 a projector is the orthogonal projection `K.starProjection` onto a subspace K of a finite-dimensional complex inner product space, and $P \leq Q$ is inclusion of subspaces; `starProjection_comp_eq_iff` proves that this agrees with $QP = P$, and `starProjection_orthogonal_eq_sub` that $I - P$ projects onto $K^\perp$. In Propositions 7.3 and 7.5 operators are complex matrices, a projector is a Hermitian idempotent matrix, a density operator is a positive semidefinite matrix of trace 1, and the Born probability is $\text{Re Tr}(\rho P)$. In Proposition 7.3 the order $P \leq Q$ is used in its algebraic form $QP = P$, and in Proposition 7.5 $|\uparrow\rangle$ and $|\downarrow\rangle$ are the standard basis vectors of $\mathbb{C}^2$. In part (2) of Proposition 7.5, “ $|\uparrow\rangle$ is the maximum-likelihood state” is proved in the form: the likelihood of $|\uparrow\rangle$ is at least that of every density operator. Propositions 7.2 and 7.6 use the same matrix model, with unitarity written $U^\dagger U = I$; Proposition 7.4 uses the lattice of subspaces of $\mathbb{C}^2$; and the entropy $S(\rho)$ is $\sum_k -\lambda_k \ln \lambda_k$ over the roots of the characteristic polynomial, which for Hermitian ρ are its eigenvalues (`vonNeumannEntropy_eq_sum_eigenvalues`).

Result	Lean declaration (namespace WangLogic)
Definition 3.2	Entails
Definition 3.4	Supported
Proposition 4.1	imp_iff_not_or
Theorem 4.2	deduction_sound
Proposition 4.4	modus_tollens
Proposition 5.1	Supported.of_cons
Theorem 5.2	Supported.mono
Proposition 5.3	falsification
Theorem 5.4	induction_not_deduction
Proposition 5.5	allWhite_supported
Proposition 6.2	m_spec
Remark 3.5	Prior.prob_and_pos_iff_supported
Definition 5.6	Prior, Confirms, Disconfirms
Proposition 5.7	Confirms.supported
Theorem 5.8	confirms_of_entails
Proposition 5.9	condProb_mono
Proposition 5.10	Disconfirms.of_refutes
Corollary 5.11	allWhite_confirmed, allWhite_condProb_mono
Definition 5.12	Normally, NormallyFrom
Theorem 5.13	Normally.of_entails, .and, .or, .cut, .cautious_mono
Proposition 5.14	normally_nonmonotone
Definition 6.3	Hierarchy
Theorem 6.4	Hierarchy.Established.of_le, .mono, .tenable, Hierarchy.Tenable.of_le, .anti
Proposition 7.1	quantum_deduction_sound
Proposition 7.3(1)	classicalProj_le_iff, classicalProj_not
Proposition 7.3(2)	classical_trace_pos_iff
Proposition 7.5(1)	QSupported.of_concat, likelihood_concat_le
Proposition 7.5(2)	spinState_supported, spinUp_maxLikelihood
Proposition 7.5(3)	spinUp_refuted
Proposition 7.5(4)	likelihood_spinState_phase
Proposition 7.2	unitary_deduction, born_unitary_conj
Proposition 7.4	quantum_not_distributive
Entropy invariance (Section 7)	vonNeumannEntropy_unitary_conj
Proposition 7.6	born_depolarize_ge, noisy_spinUp_not_refuted

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